

Comp_my_vs_Keld.doc

[1]. Sov. Phys. JETP 20, 1307 (1964), / ionization in the field of a strong electromagnetic wave. (Л.В. Келдыш Ионизация в поле сильной электромагнитной волны ЖЭТФ 1964, т.47, в. 5(11), с.1945-1957. (L.V.Keldysh Zh.Eksp.Teor.Fiz.47, 1945 (1964)).

[2]. J.R. Gulley Ultrafast laser-induced damage and the influence of spectral effects//Optical Engineering 2012, Vol. 51(12), 121805 8 pages).

Definitions.

A). $w_0 \text{ s}^{-1}$ – atom ionization rate calculated numerically under following assumptions:

1. Hydrogen-like ion with ground state ionization energy $E_{g0} = m_e \alpha^2 / (2\hbar^2)$, $\alpha = Ze^2$, (Z is ion charge) is assumed.
2. Only ground state with wave function $\psi_0 = (\pi a_a^3)^{-1/2} e^{iE_{g0}t - r/a_a}$ ($a_a = \hbar^2 / (m_e \alpha)$, $E_{g0} a_a^2 = \hbar^2 / (2m_e)$) and states of continuum spectrum are involved in the interaction. Coulomb potential for states of continuum spectrum is not considered in Schrodinger equation. Ground state is assumed to be unperturbed (low degree of ionization)
3. Ionization over long time $t \gg \omega^{-1}$ is studied.

Equations (8), (9) from [1] are obtained as a result of assumptions above. (However, [1] apparently has a mistake: correct result should be 2 times higher than in [1]. It was mentioned by other authors as well) In this work, integrals (8) and (9) are calculated numerically. Thus, w_0 is obtained.

B). $w_s \text{ s}^{-1}$ is ionization rate w_0 calculated approximately by saddle-point method. (eq. (16)-(19) in [1]).

C). $w_Q \text{ s}^{-1}$ is ionization rate w_s , considering Coulomb interaction for continuum states (eq. (1) and (21) in [1]).

D). $W_C \text{ s}^{-1} \text{ sm}^{-3}$ is ionization rate for a crystal (eq. (37)-(39) in [1]). Unlike for atom, W_C is calculated per volume unit, not per atom, in $[\text{s}^{-1}]$.

Case of atom. Comparison of w_0 and w_s .

Assumption of Coulomb interaction.

After considering lost 2 in (8), (9) and correcting an obvious misprint in (16) ($\sqrt{2I_0/\hbar\omega}$ instead of $\sqrt{2I_0/\hbar}$ should be written) the formula for w_s is as follows:

$$w_s = 2\omega\sqrt{2x_0}\left(\frac{\gamma}{\sqrt{1+\gamma^2}}\right)^{3/2} \exp\left[-2x\left(\ln\left(\gamma+\sqrt{1+\gamma^2}\right)-\gamma\frac{\sqrt{1+\gamma^2}}{1+2\gamma^2}\right)\right] \cdot \exp\left[-2(\langle x+1\rangle-x)\left(\ln\left(\gamma+\sqrt{1+\gamma^2}\right)-\frac{\gamma}{\sqrt{1+\gamma^2}}\right)\right] \cdot \sum_{n=0}^{\infty} \Phi(z_n) \exp\left[-2n\left(\ln\left(\gamma+\sqrt{1+\gamma^2}\right)-\frac{\gamma}{\sqrt{1+\gamma^2}}\right)\right] \quad (1)$$

$$x_0 = \frac{I_0}{\hbar\omega}$$

$$\varepsilon = E/E_*, \quad E_* = \alpha e^{-1}(2mE_{g0})^{1/2} \quad (2)$$

$$x = x_0(1 + \varepsilon^2/2)$$

$$z_n = \left[\frac{2\gamma}{\sqrt{1+\gamma^2}} (\langle x+1 \rangle - x + n) \right]^{1/2}$$

$$\gamma = \varepsilon^{-1}$$

$$\Phi(z) = \int_0^z e^{y^2 - z^2} dy \quad (\text{for } z \rightarrow \infty \quad \Phi \approx (2z)^{-1})$$

or

$$w_s = 2\omega\sqrt{2x_0}\left(\frac{\gamma}{\sqrt{1+\gamma^2}}\right)^{3/2} \left(\frac{\varepsilon}{2}\right)^{2\langle x+1 \rangle} \exp\left[\frac{-2x\gamma^3}{(1+2\gamma^2)\sqrt{1+\gamma^2}} - 2\langle x+1 \rangle \left(\ln\left(\frac{\gamma+\sqrt{1+\gamma^2}}{2\gamma}\right) - \frac{\gamma}{\sqrt{1+\gamma^2}}\right)\right]$$

$$\cdot \sum_{n=0}^{\infty} \Phi(z_n) \left(\frac{\varepsilon}{2}\right)^{2n} \exp\left[-2n\left(\ln\left(\frac{\gamma+\sqrt{1+\gamma^2}}{2\gamma}\right) - \frac{\gamma}{\sqrt{1+\gamma^2}}\right)\right] \quad (3)$$

The results can be found in my programs. Coefficients are not tracked during calculations, since it is assumed that they must be selected correspondingly to particular experiment. In order to express w_0 in terms of fs^{-1} , the value of w_0 obtained after my calculations must be multiplied by

$$W_{p10}[\text{fs}^{-1}] = \frac{128E_{g0}}{\pi\hbar} = 61.9E_{g0}[\text{eV}].$$

The comparison of w_0 and formulas for w_s obtained with saddle-point method in [1] for the wavelength 400 and 1030 nm, $E_{g0}=9$ eV is shown in figure 1. The x-axis represents value $\log_{10}(5.46 \cdot 10^{13}(E^2/E_*^2))$ – logarithm of intensity in terms of W/cm^2 . Intensity

$I[\text{W}/\text{cm}^2] = \frac{cn_s E_*^2}{8\pi} \varepsilon^2 = 7.53 \cdot 10^{13} \cdot n_s m \varepsilon^2 = 5.46 \cdot 10^{13} \varepsilon^2$, where $n=1.45$ – is the refractive index, $m=1/2$ – is reduced mass expressed in the mass of an electron in vacuum. $E_* = \omega \varepsilon^{-1} (2mE_{g0})^{1/2}$.

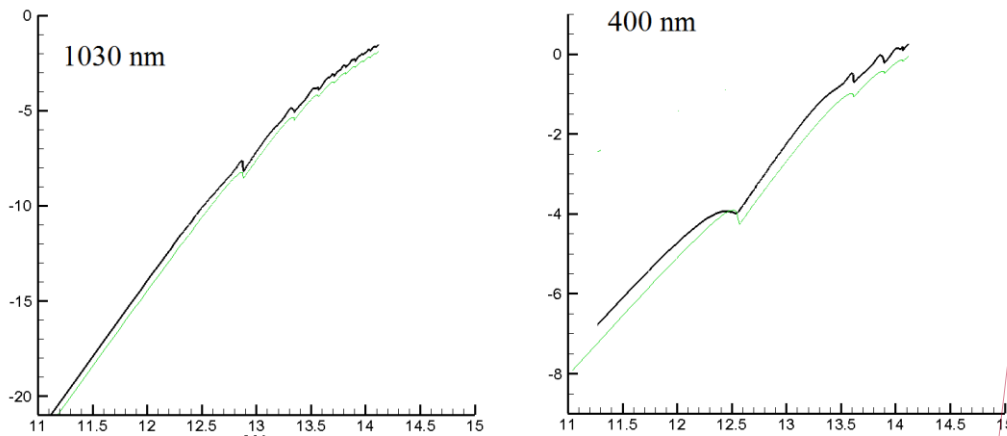


Fig. 1 $\log(w_0 [\text{fs}^{-1}])$ – black curve, $\log(w_s [\text{fs}^{-1}])$ – green curve, x-axis: $\log(I[\text{W}/\text{cm}^2])$

The case for 800 nm is introduced more detailed in figure 2. Differences are not noticeable in logarithmic scale, especially in the regions of ε far from the points corresponding to vanishing of n-photon processes. Therefore values $w_0(E_*^2/E^2)^6$ and $w_s(E_*^2/E^2)^6$ are shown on the upper side of figure 2. Here differences between Keldysh's and ~~mine-my~~ results become more obvious. On fig 2a the value $\Delta = 2(w_0 - w_s)/(w_0 + w_s)$ is also shown. It is necessary to take into account, that Keldysh theory has only logarithmic accuracy.

Commented [TD1]: I understand that I have to apply this formula to use the single-color Keldysh rates files you provided me. I also understand I should multiply them with the density of the solid. The single-color rates you sent are expressed in $1/\text{fs}$, yes?

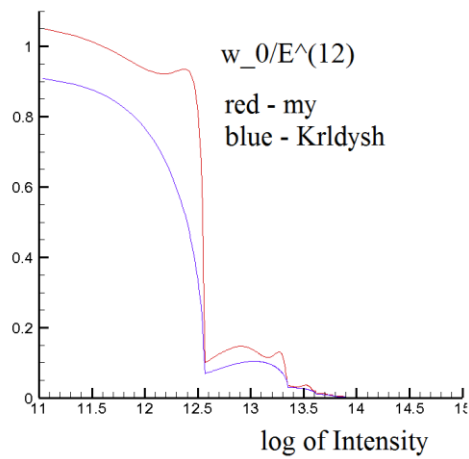
Commented [TD2]: I understand that agreement with paper of Gruzdev can be obtained only if using optical refractive index everywhere (due to pulse compression in matter). This enables to repeat 1 to 1 the results of Gruzdev. I agree to use it.

Commented [TD3]: Could you provide the results at 800 nm for Si (gap 2.56 eV. And effective mass 0.2226)? Then I could introduce them in my comparison with TDDFT model.

Commented [TD4]: Does it refer to the 6 photon-excitation in SiO_2 ? Optical index n may be missing then.

Commented [TD5]: Does it refer to some convergence rate?

800 nm, 9 eV



w_0 in logarithmic coordinate
green - Keldysh, black - my.
Red for the case Keldysh 2 times less.

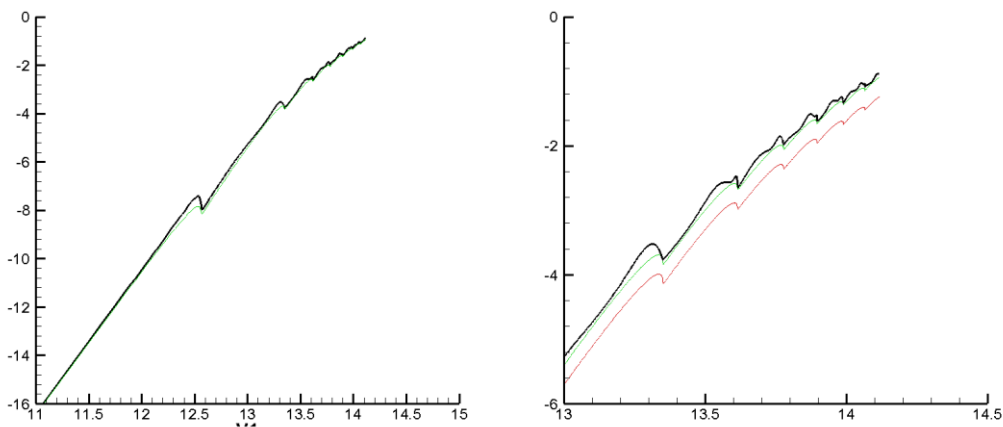


Fig. 2 Top plot: values $w_0(E_*^2/E^2)^6 \text{ fs}^{-1}$ (red) and $w_s(E_*^2/E^2)^6 \text{ fs}^{-1}$ (black)
Bottom plots: $\log(w_0 [\text{fs}^{-1}])$ (black) and $\log(w_s [\text{fs}^{-1}])$ (green), x-axis:
 $\log(I[\text{W}/\text{cm}^2])$.

Commented [TD6]: 6 photons: I understand that it must be for SiO2, yes?

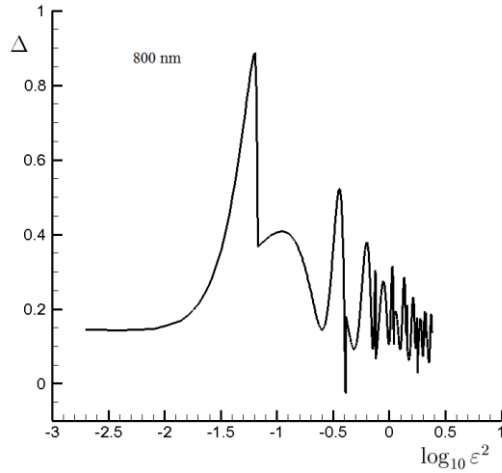


Fig. 2a. The value Δ for 800 nm, $E_{g0}=9$ eV.

I will make a number of important observations.

1. Keldysh's article specifies that developed theory is only applicable for $E_{g0}/\hbar\omega \gg 1$. In our case this value is ≈ 6 (for fused silica) or less (for silicon).
2. In Keldysh's and this work, matrix element of **hydrogen-like ion** is used. The real matrix element may be different. The statement about weak dependency of the result on the matrix element in Keldysh's article is technically connected with use of saddle-point method, which will be used is case $E_{g0}/\hbar\omega \gg 1$. In our case this inequality is often not fulfilled.
3. As stated before, developed theory doesn't consider Coulomb interaction. Keldysh states that for taking this effect into account it is sufficient to multiply obtained result by

$$A(I_0/\hbar\omega)\left(\gamma/\sqrt{1+\gamma^2}\right) \quad (4)$$

Here A is a coefficient in the order of 1. As a result in Keldysh's work the final formulas (1) and (21) are obtained. The first is as follows:

$$w_Q = A\omega x_0^{3/2} \left(\frac{\gamma}{\sqrt{1+\gamma^2}} \right)^{5/2} \exp \left[-2x \left(\ln \left(\gamma + \sqrt{1+\gamma^2} \right) - \gamma \frac{\sqrt{1+\gamma^2}}{1+\gamma^2} \right) \right].$$

Commented [TD7]: Could I ask which formula this refers to? It would be interesting to discuss this topic further (for future). For example, we could consider using the existing pseudo-potentials to prepare material dependent Keldysh rates, not only as function of the gap energy, but also as function of the atomic potentials themselves.

Commented [TD8]: Coulomb interaction may be very difficult to account for. I don't think it would be so simple to account for it.

$$\cdot \exp \left[-2(\langle x+1 \rangle - x) \left(\ln(\gamma + \sqrt{1+\gamma^2}) - \frac{\gamma}{\sqrt{1+\gamma^2}} \right) \right] \cdot \sum_{n=0}^{\infty} \Phi(z_n) \exp \left[-2n \left(\ln(\gamma + \sqrt{1+\gamma^2}) - \frac{\gamma}{\sqrt{1+\gamma^2}} \right) \right] \quad (5)$$

In cases of our interest $\gamma \geq 1$, the value of (4) is ≈ 5 and is weakly depending on field.

Limited by low fields ($\gamma \gg 1$, multiphoton ionization) for atom from (5) we have

$$w_0 \approx 2\omega\sqrt{2x_0} \left(\frac{\varepsilon}{2} \right)^{2\langle x_0+1 \rangle} \exp[2\langle x_0+1 \rangle - x_0] \Phi(z_*), \quad (6)$$

$$z_* = [2(\langle x_0+1 \rangle - x_0)]^{1/2}$$

And considering a factor in the order of magnitude of (4)

$$w_Q \approx A\omega x_0^{3/2} \left(\frac{\varepsilon}{2} \right)^{2\langle x_0+1 \rangle} \exp[2\langle x_0+1 \rangle - x_0] \Phi(z_*) \quad (7)$$

Generally, developed theory is very approximate. It is noted in Keldysh's work that "Pre-exponential factor... needs to be clarified, however all dependencies are defined by exponents. Thus, coefficient of obtained formulas can be taken as an adjustable parameter, which is determined experimentally."

Commented [TD9]: I can now prepare adjustment of the fitting parameter on the TDDFT simulation results.

We shall notice that factor (4) depends on frequency and in case of a two color pulse is difficult to determine. However, in a two color pulse frequencies don't show bigger discrepancies than a factor of 2. That means for different ratios between wave amplitudes contained in a dichromatic pulse, the coefficient can vary only by amount in the order of unity. However, typically in two color pulses frequencies differ not more than of a factor of two. Such details can be neglected considering the proximity of obtained formulas.

Case of crystal

References to elliptical integrals.

$$E(k) = \int_0^{\pi/2} \sqrt{1-k^2 \sin^2 \varphi} d\varphi = \int_0^1 \frac{\sqrt{1-k^2 t^2}}{\sqrt{1-t^2}} dt = \frac{\pi}{2} \left[1 - \left(\frac{1}{2}\right)^2 k^2 - \left(\frac{1 \cdot 3}{2 \cdot 4}\right)^2 \frac{k^4}{3} - \left(\frac{(2m-1)!!}{(2m)!!}\right) \frac{k^{2m}}{2m-1} \right]$$

,

$$K(k) = \int_0^{\pi/2} \frac{d\varphi}{\sqrt{1-k^2 \sin^2 \varphi}} = \frac{\pi}{2} \left[1 + \left(\frac{1}{2}\right)^2 k^2 + \left(\frac{1 \cdot 3}{2 \cdot 4}\right)^2 k^4 + \left(\frac{(2m-1)!!}{(2m)!!}\right)^2 k^{2m} \right].$$

$$\lim_{x \rightarrow 1} \left(K(x) - \frac{1}{2} \ln \left(\frac{16}{1-x^2} \right) \right) = 0 \text{ или } \lim_{x \rightarrow 0} K(\sqrt{1-x^2}) = \frac{1}{2} \ln \left(\frac{16}{x^2} \right)$$

Through Legendre formula

$$K(x)E(\sqrt{1-x^2}) + E(x)K(\sqrt{1-x^2}) = \frac{\pi}{2} + K(x)K(\sqrt{1-x^2})$$

$$\text{We get } \lim_{x \rightarrow 0} E(\sqrt{1-x^2}) = 1 + \frac{x^2}{4} \ln \left(\frac{16}{x^2} \right)$$

$K(k)$ can be calculated as follows. Let's consider the sequences $a_{n+1} = \frac{a_n + b_n}{2}$, $b_{n+1} = \sqrt{a_n b_n}$ for starting values $a_0 = k$, $b_0 = \sqrt{1-k^2}$. Then for such n , when $a_n = b_n$ we have $K(k) = \frac{\pi}{2a_n}$.

In [1] semiconductor (crystal) deviates from the case of atom discussed above by relation between impulse and energy. For crystals this relation is:

$$\varepsilon(p) = E_{g0} \sqrt{1 + p^2 / (mE_{g0})} \quad (6)$$

In the case of **small pulse amplitudes**, we have

$$\varepsilon(p) \approx E_{g0} + p^2 / (2m) - p^4 / (8m^2 E_{g0}) \quad (7)$$

Moreover, electrons interact with the lattice (with many atoms) in a crystal. Therefore, unlike in the case of atom, ionization probability w_c **cm^{-3}** per volume unit rises in crystals. Qualitatively dependency of w_c on pulse energy holds as showed at fig. 1,2.

In case of crystal, for small amplitudes according to Keldysh: (formula (39) of Keldysh is misprinted. Instead of $/2KE$, $/4KE$ had to be printed. It is corrected in Gulley paper. To obtain the formula (41) of Keldysh paper must be 4 instead of 2 also. This had been checked^ coincidence with Gulley)

$$w_c = \frac{9}{2\pi} \omega \left(\frac{m\omega}{\hbar} \right)^{3/2} \left(\frac{\varepsilon}{2\sqrt{2}} \right)^{2(x_0+1)} \exp[2(x_0+1)] \Phi(z_*) \quad (8)$$

Commented [TD10]: Beyond MPI regime, this approximation is broken.

Commented [TD11]: Is it $\text{cm}^{-3}/\text{s}^{-1}$? ☺

Commented [TD12]: But Gruzdev does not account for it. However, I may have found that Gulley is using a series that does not converge. Hence, I don't use his paper.

Plot from Gulley's work and results of my own calculations multiplied by density $\rho_s = 10^{20} \text{ cm}^{-3}$ are shown in fig. 3. Considering approximation of formulas, very good correspondence can be seen. Note that the transition from n-photon to n+1-photon ionization occurs at higher intensities in case of atoms than it is the case for crystals, as must be according to (7). Note also that in case of crystal dependency of energy on pulse can differ from (6) (Gruzdev). This fact introduces additional arbitraries to formulas.

We emphasize that value $\rho_s w_0$ can be interpreted as probability of one atom ionization multiplied by an effective density of atoms. The effective density of atoms is not directly connected to real density of atoms in a crystal. This is because the behavior of an electron in a crystal is determined by the interaction with the entire lattice (many atoms). It is impossible to single out one atom.

Note that fused silica is not a crystal. Thus, technically speaking crystal theory can't be applied. On the other hand, density in fused silica is very high. Hence, atoms can't be considered as isolated from each other.

Conclusions

The main conclusion of this work is that for γ values in the range of interest and not exceeding 2÷3, the results of the different models (atom and saddle-point method for integral value calculation (Keldysh) and precise integral calculation (current work, Keldysh's theory for crystal)) are very close. Considering multiple assumptions, the differences between results are insignificant. Thus, the simpler model for photoionisation can be used, for example for further development of photoionisation theory for two color pulses. Wherein a coefficient, which is determined from comparison with experimental results or from calculations based on initial principles, must be introduced. The electron's mass must be considered as adjustable parameter as well.

Ultrafast laser-induced damage and the influence of spectral effects

red - my keen * $10^{16}(\text{fs} \rightarrow \text{s}) * 10^{20}$ so $\rho = 10^{20} \text{ sm}^{-3}$

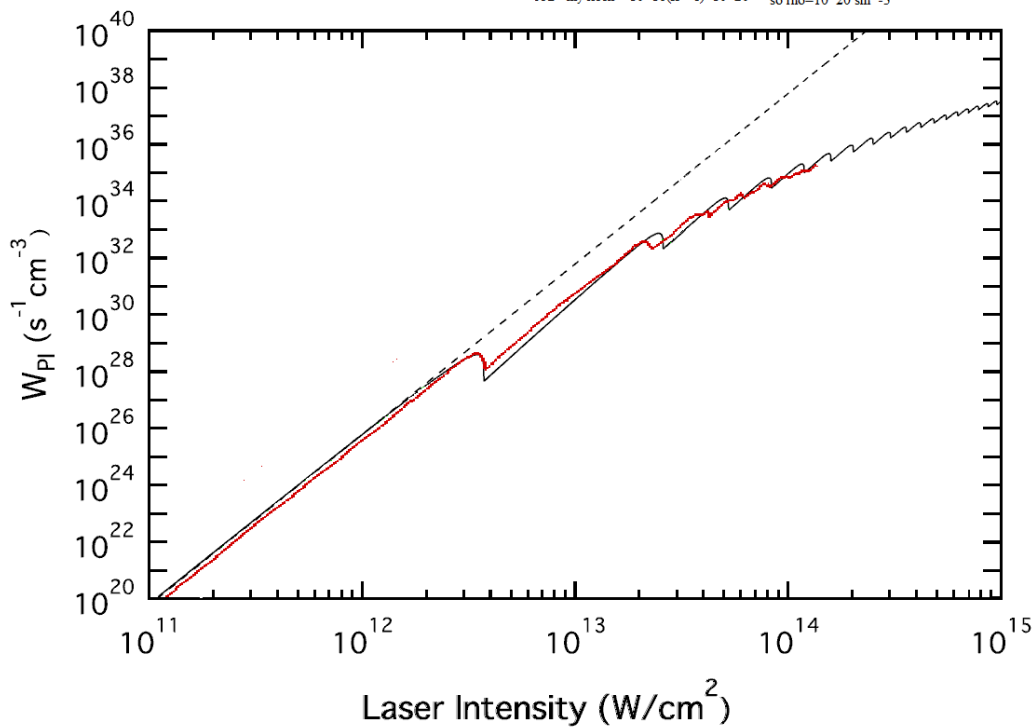


Рис. 3 Red curve: $\log(w_0 \rho_e)$, black curve: $\log(W_C)$, from Gulley's article

The radiation wavelength is much longer than size of the atom. Thus, the electrical field depends only on time and is defined as follows:

$$E = E_1(t)\mathbf{e}_1 \cos(\omega_1 t) + E_2(t)\mathbf{e}_2 \cos(\omega_2 t + \varphi_2),$$

Where E_1, E_2 states for first and second harmonic, which are polarized along constant vectors $\mathbf{e}_1, \mathbf{e}_2$. Some of following calculations can be performed for time-depended amplitudes. Thus, time dependence of E_1, E_2 is still considered. Harmonics can be shifted by phase φ_2 . The main results are obtained for $E_{1,2} = \text{const}$. In [1] the case of $E_1 = \text{const}$, $E_2 = 0$ is considered.

Commented [TD13]: I have rather worked with the paper of Gruwdev 2014. I could repeat his numerical calculations 1 to 1. But Gulley seems to have made a mistake in writing his formulas in the paper, as the sum he indicated does not converge.

Commented [TD14]: Do you mean $E(t) = E_1 \mathbf{e}_1 \cos(\dots) + E_2 \mathbf{e}_2 \cos(\dots)$ instead?

To make a clear overview, all values are normalized as follows. Distance will be measured in terms of $a_0 = \hbar^2 / (m_e \alpha)$ and time in $\hbar / E_{g0}$, where $E_{g0} = \frac{m_e \alpha^2}{2\hbar^2}$ is the ground state energy; frequency will be measured in $E_{g0} / \hbar$, electrical field in Keldysh's units $E_{*1,2} = \omega_{1,2} (2m_e E_{g0})^{1/2} e^{-1}$. Thus, the electrical field of each component is normalized on its own value. Then the dimensionless quantities $\varepsilon_{1,2} = E_{1,2} / E_{*1,2}$ will be used instead of electrical fields. Hence, the Schrödinger equation for a continuous spectrum function ψ takes the form:

Commented [TD15]: What is alpha? Is it the fine structure constant? Did you use atomic units or CGS here? In atomic units, alpha is the speed of light in vacuum.

$$\psi = e^{i\mathbf{p}\mathbf{r}/\hbar}, \quad -i\hbar \psi = \hat{p}\psi$$

$$\psi = \int e^{i(\mathbf{p}_1 - \mathbf{p}_2)\mathbf{r}/\hbar} d^3r = (2\pi\hbar)^3 \delta(\mathbf{p}_1 - \mathbf{p}_2)$$

$$\psi(\mathbf{r}) = \int a(\mathbf{p}) e^{i\mathbf{p}\mathbf{r}/\hbar} \frac{d^3p}{(2\pi\hbar)^3}, \quad a(\mathbf{p}) = \int \psi(\mathbf{r}) e^{-i\mathbf{p}\mathbf{r}/\hbar} d^3r$$

$$\mathbf{r} = i\hbar \frac{\partial}{\partial \mathbf{p}}$$

$$\int |\psi|^2 d^3r = \int |a(\mathbf{p})|^2 \frac{d^3p}{(2\pi\hbar)^3}$$

$$i\hbar \psi_t = -\frac{\hbar^2}{2m} \Delta \psi + U\psi, \quad E = -\nabla U, \quad U = eE(\mathbf{er}), \quad e > 0$$

Commented [TD16]: Does E(er) refers to dipolar approximation of the electric field?

$$i\hbar \psi_t - e(E_1(t)(\mathbf{e}_1\mathbf{r})\cos(\omega_1 t) + E_2(t)(\mathbf{e}_2\mathbf{r})\cos(\omega_2 t + \varphi_2)) \left(\psi + \frac{1}{\sqrt{\pi a_a^3}} e^{-r/a_a + iE_{g0}t/\hbar} \right) = -\frac{\hbar^2}{2m} \Delta \psi$$

$$\frac{\partial a_p}{\partial t} - e(E_1(t)(\mathbf{e}_1\nabla_p)\cos(\omega_1 t) + E_2(t)(\mathbf{e}_2\nabla_p)\cos(\omega_2 t + \varphi_2))a_p = -i\frac{p^2}{2m\hbar}a_p -$$

$$-\frac{ie^{iE_{g0}t/\hbar}}{\hbar} \int e^{-i\mathbf{p}\mathbf{r}/\hbar} e(E_1(t)(\mathbf{e}_1\mathbf{r})\cos(\omega_1 t) + E_2(t)(\mathbf{e}_2\mathbf{r})\cos(\omega_2 t + \varphi_2)) \frac{e^{-r/a_a}}{\sqrt{\pi a_a^3}} d^3r$$

$$-\frac{1}{\hbar} \int e^{-i\mathbf{p}\mathbf{r}/\hbar} eE(\mathbf{er}) \frac{e^{-r/a_a}}{\sqrt{\pi a_a^3}} d^3r = -8\sqrt{\pi a_a^3} eE(\mathbf{e}\nabla_p)(1 + p^2 a_a^2 / \hbar^2)^{-2} = \frac{32\sqrt{\pi a_a^3} eE(\mathbf{ep}) a_a^2}{(1 + p^2 a_a^2 / \hbar^2)^3 \hbar^2} =,$$

$$= \frac{32\sqrt{\pi a_a^3} E / E_*(\mathbf{ep})}{(1 + p^2 a_a^2 / \hbar^2)^3} \frac{a_a}{\hbar} \omega = 32\sqrt{\pi a_a^3} \frac{\omega(E / E_*)(\mathbf{e}(\mathbf{p}a_a / \hbar))}{(1 + p^2 a_a^2 / \hbar^2)^3},$$

Considering $E_* = \omega e^{-1}(2mE_{g0})^{1/2}$ and $a_a^2 E_{g0} = \frac{\hbar^2}{2m}$.

$$\begin{aligned} \frac{\partial a_p}{\partial t} - e(E_1(t)(\mathbf{e}_1 \nabla_p) \cos(\omega_1 t) + E_2(t)(\mathbf{e}_2 \nabla_p) \cos(\omega_2 t + \varphi_2)) a_p = -i \frac{p^2}{2m\hbar} a_p + \\ + ie^{iE_{g0}t/\hbar} 32\sqrt{\pi a_a^3} \frac{\omega_1(E_1 / E_{*1})(\mathbf{e}_1(\mathbf{p}a_a / \hbar) \cos(\omega_1 t) + \omega_2(E_2 / E_{*1})(\mathbf{e}_2(\mathbf{p}a_a / \hbar) \cos(\omega_2 t + \varphi_2))}{(1 + p^2 a_a^2 / \hbar^2)^3} \end{aligned}$$

We define $\varepsilon_{1,2} = E_{1,2} / E_{*1,2}$, $\{\omega\varepsilon, \mathbf{g}\} = \omega_1\varepsilon_1(t)(\mathbf{e}_1\mathbf{g})\cos(\omega_1 t) + \omega_2\varepsilon_2(t)(\mathbf{e}_2\mathbf{g})\cos(\omega_2 t + \varphi_2)$

$$\{\mathbf{F}, \mathbf{g}\} = (\mathbf{F}_1\mathbf{g})\cos(\omega_1 t) + (\mathbf{F}_2\mathbf{g})\cos(\omega_2 t + \varphi_2)$$

For constant field amplitude characteristics $\mathbf{P} = \xi - \mathbf{q}(t)$,
 $\mathbf{q} = \frac{eE_1}{\omega_1} \mathbf{e}_1 \sin(\omega_1 t) + \frac{eE_2}{\omega_2} \mathbf{e}_2 \sin(\omega_2 t + \varphi_2) = \frac{\hbar}{a_a} (\varepsilon_1 \mathbf{e}_1 \sin(\omega_1 t) + \varepsilon_2 \mathbf{e}_2 \sin(\omega_2 t + \varphi_2))$ and solution is:

$$a_\xi = 32i\sqrt{\pi a_a^3} e^{-\frac{i}{2m\hbar} \int_0^t (\xi - \mathbf{q}(\tau))^2 d\tau} \int_0^t \frac{\{\omega\varepsilon(\tau), (\xi - \mathbf{q}(\tau))a_a / \hbar\}}{(1 + (\xi - \mathbf{q}(\tau))^2 a_a^2 / \hbar^2)^3} e^{\frac{i}{\hbar} E_{g0}\tau + \frac{i}{2m\hbar} \int_0^\tau (\xi - \mathbf{q}(\nu))^2 d\nu} d\tau$$

In Keldysh, characteristics have $+$. Obviously, $e < 0$.

$$\frac{\sin^2(\alpha t)}{\pi \alpha^2 t} = \delta(\alpha)$$

Commented [TD17]: Do you mean that Keldysh's paper uses a negative charge, but you have used a positive one, yes?

Determination of W_{PI}

$$1. \quad w = \lim_{t \rightarrow \infty} \left(\frac{\int |\psi|^2 d^3 r}{t} \right) \text{ found by myself. If}$$

$$|a^2| = w_0 t + (\text{типа } \sin(\omega t)), \quad (1)$$

Then $w = w_0$ as expected.

Commented [TD18]: I would propose $\lim_{t \rightarrow \infty} \int_0^t |\psi|^2 / \tau$, where τ is pulse duration. Otherwise, w may be 0, as total density is a finite number.

Commented [TD19]: Kind of ... OK ☺

$$2. \quad w = \lim_{t \rightarrow \infty} \left(\frac{\partial \int |\psi|^2 d^3 r}{\partial t} \right) \equiv 2 \lim_{t \rightarrow \infty} \operatorname{Re} \int a \partial a^* / \partial t d^3 r - \text{according to Keldysh. It seems}$$

that multiplier **2** was abandoned in Keldysh's work. In case (1) $w = w_0 + (\text{terms like } \cos(\omega t))$, thus oscillating! Keldysh expanded in Fourier series without electric field $\cos(\omega t)$. This is why $L_{n+1} + L_{n-1}$ is present there.

$$\text{We will keep 1st definition. } w = \lim_{t \rightarrow \infty} \left(\frac{\int |a_p|^2 d^3 p}{(2\pi\hbar)^3 t} \right) \equiv \lim_{t \rightarrow \infty} \left(\frac{\int |a_\xi|^2 d^3 \xi}{(2\pi\hbar)^3 t} \right)$$

$$w = \frac{2^{10} \pi a_a^3}{(2\pi\hbar)^3} \lim_{t \rightarrow \infty} \int t^{-1} \left| \int_0^t \frac{\{\omega \varepsilon(\tau), (\xi - \mathbf{q}(\tau)) a_a / \hbar\}}{(1 + (\xi - \mathbf{q}(\tau))^2 a_a^2 / \hbar^2)^3} e^{\frac{i}{\hbar} E_{g0} \tau + \frac{i}{2m\hbar} \int_0^\tau (\xi - \mathbf{q}(\nu))^2 d\nu} d\tau \right|^2 d^3 \xi =$$

$$= \frac{2^{10} \pi}{(2\pi)^3} \lim_{t \rightarrow \infty} \int t^{-1} \left| \int_0^t \frac{\{\omega \varepsilon(\tau), (\xi - \mathbf{q}(\tau) a_a / \hbar)\}}{(1 + (\xi - \mathbf{q}(\tau) a_a / \hbar)^2)^3} e^{\frac{i}{\hbar} E_{g0} \tau + \frac{i}{2m\hbar} \int_0^\tau (\xi - \mathbf{q}(\nu))^2 d\nu} d\tau \right|^2 d^3 \xi$$

Commented [TD20]: If you agree with my previous comment, we may update the equations of this page.

Commented [TD21]: Why 2^{10} ? What is its origin?

Commented [TD22]: $t^{\wedge-1}$ after the integral may be a $t_{\text{pulse}}^{\wedge-1}$

After $\bar{\mathbf{q}} = \varepsilon_1 \mathbf{e}_1 \sin(\omega_1 t) + \varepsilon_2 \mathbf{e}_2 \sin(\omega_2 t + \varphi_2)$ is introduced, and time is defined in terms $\hbar / E_{g0}$, and frequency in $E_{g0} / \hbar$ we have

$$w = \frac{2^{10} \pi E_{g0}}{(2\pi)^3 \hbar} \lim_{t \rightarrow \infty} \int t^{-1} \left| \int_0^t \frac{\{\omega \varepsilon(\tau), (\xi - \bar{\mathbf{q}}(\tau))\}}{(1 + (\xi - \bar{\mathbf{q}}(\tau))^2)^3} e^{i\tau + i \int_0^\tau (\xi - \bar{\mathbf{q}}(\nu))^2 d\nu} d\tau \right|^2 d^3 \xi$$

Or

$$w = \frac{2^{10} \pi E_{g0}}{(2\pi)^3 \hbar} \lim_{t \rightarrow \infty} \int t^{-1} |b_\xi|^2 d^3 \xi$$

$$b_\xi = \int_0^t \left[\omega_1 \varepsilon_1(\tau) \cos(\omega_1 \tau) (\mathbf{e}_1 \cdot (\xi - \bar{\mathbf{q}}(\tau))) + \omega_2 \varepsilon_2(\tau) \cos(\omega_2 \tau + \varphi_2) (\mathbf{e}_2 \cdot (\xi - \bar{\mathbf{q}}(\tau))) \right] \times$$

$$\times \frac{\exp \left(i\tau + i \int_0^\tau (\xi - \mathbf{q}(\nu))^2 d\nu \right)}{(1 + (\xi - \mathbf{q}(\tau))^2)^3} d\tau$$

$$b_\xi = \int_0^t e^{i(1+\xi^2+\varepsilon_1^2/2+\varepsilon_2^2/2)\tau} f e^{i\eta(\tau)} d\tau$$

$f e^{i\eta(\tau)}$ will be expanded into Fourier series. Then

$$b_\xi = \sum_{n,m} f_{nm} \int_0^t e^{i\alpha\tau - i\omega_1 n\tau - i\omega_2 m\tau} d\tau = \sum_{n,m} f_{nm} \frac{e^{i\alpha\tau - i\omega_1 n\tau - i\omega_2 m\tau} - 1}{i(\alpha - \omega_1 n - \omega_2 m)},$$

$$\alpha = 1 + \xi^2 + \varepsilon_1^2 / 2 + \varepsilon_2^2 / 2$$

$$w = \frac{2^{10} \pi E_{g0}}{(2\pi)^3 \hbar} \lim_{t \rightarrow \infty} \int t^{-1} \sum_{n,m,n',m'} f_{nm} f_{n'm'}^* \frac{(e^{i\alpha\tau - i\omega_1 n \tau - i\omega_2 m \tau} - 1)(e^{-i\alpha\tau + i\omega_1 n' \tau + i\omega_2 m' \tau} - 1)}{(\alpha - \omega_1 n - \omega_2 m)(\alpha - \omega_1 n' - \omega_2 m')} d^3 \xi$$

If $n' = n$ and $m' = m$ then

$$\frac{(e^{i\alpha\tau - i\omega_1 n \tau - i\omega_2 m \tau} - 1)(e^{-i\alpha\tau + i\omega_1 n \tau + i\omega_2 m \tau} - 1)}{(\alpha - \omega_1 n - \omega_2 m)(\alpha - \omega_1 n' - \omega_2 m')} = \frac{4 \sin^2((\alpha - \omega_1 n - \omega_2 m) / 2)}{(\alpha - \omega_1 n - \omega_2 m)^2} \rightarrow$$

$$\rightarrow \pi t \delta((\alpha - \omega_1 n - \omega_2 m) / 2) = 2\pi t \delta(\alpha - \omega_1 n - \omega_2 m)$$

Terms with $n' \neq n$ and/or $m' \neq m$ contribute to w as

$$\lim_{t \rightarrow \infty} t^{-1} \int \theta(\xi) \frac{(e^{i\xi^2 \tau - i\beta \tau} - 1)(e^{-i\xi'^2 \tau + i\beta' \tau} - 1)}{(\xi^2 - \beta)(\xi'^2 - \beta')} d^3 \xi = \lim_{t \rightarrow \infty} t^{-1} \int \xi \theta(\xi) \frac{(e^{i\xi^2 \tau - i\beta \tau} - 1)(e^{-i\xi'^2 \tau + i\beta' \tau} - 1)}{(\xi^2 - \beta)(\xi'^2 - \beta')} d\xi^2 d\Omega_\xi$$

The integrand has no singularities. Hence, the integral will be calculated as principal value after expanding brackets in numerator and representing denominator as $\frac{1}{(\xi^2 - \beta)(\xi'^2 - \beta')} = \frac{1}{\beta - \beta'} \left(\frac{1}{\xi^2 - \beta} - \frac{1}{\xi'^2 - \beta'} \right)$. Principal value of integral $1/x$ is finite. Principal value of $\int \mathcal{G}(x)(e^{ixt} - 1) / x dx = \int \mathcal{G}(y/t)(e^{iy} - 1) / y dy = \mathcal{G}(0) \int (e^{iy} - 1) / y dy$ doesn't increase with time. Thus, terms with $n' \neq n$ and/or $m' \neq m$ contribute to w with a damping term.

Thus,

$$w = \frac{2^8 E_{g0}}{\pi \hbar} \int \sum_{n,m} |f_{nm}|^2 \delta(1 + \xi^2 + \varepsilon_1^2 / 2 + \varepsilon_2^2 / 2 - \omega_1 n - \omega_2 m) d^3 \xi \approx$$

$$\approx 1.24 \cdot 10^{15} E_{g0} [\text{eV}] \int \sum_{n,m} |f_{nm}|^2 \delta(1 + \xi^2 + \varepsilon_1^2 / 2 + \varepsilon_2^2 / 2 - \omega_1 n - \omega_2 m) d^3 \xi \text{ ceK}^{-1}.$$

$$\int |f_{m,n}|^2 \delta(1 + \xi^2 + \varepsilon_1^2 / 2 + \varepsilon_2^2 / 2 - \omega_1 n - \omega_2 m) d^3 \xi = \quad (14)$$

$$= \int |f_{m,n}(\xi_\perp^2, \xi_z, \varphi)|^2 \delta(\xi_\perp^2 - (\xi_{m,n}^2 - \xi_z^2)) \xi_\perp d\xi_\perp d\xi_z d\varphi = W_{m,n} / 2 \quad (15)$$

$$W_{m,n} = \int_{-\xi_{m,n}}^{\xi_{m,n}} \int_0^{2\pi} |f_{m,n}(\xi_{m,n}^2 - \xi_z^2, \xi_z, \varphi)|^2 d\xi_z d\varphi = 2 \int_{-\xi_{m,n}}^{\xi_{m,n}} \int_{-\pi/2}^{\pi/2} |f_{m,n}(\xi_{m,n}^2 - \xi_z^2, \xi_z, \varphi)|^2 d\xi_z d\varphi$$

For internal demonstration. In programs (for example keldu5.for) unnormalized ionization speed equals to $\sum_{n,m} W_{n,m}$ (multiplied by 2 if calculated from $(-\pi/2)$ to $\pi/2$).

Then

$$W_{PI} = W_{PI0} \sum_{n,m=-\infty}^{\infty} W_{m,n} \quad (16)$$

Considering numerical coefficients in (5) $W_{PI0}[\text{s}^{-1}] = \frac{128E_{g0}}{\pi\hbar} = 6.19 \cdot 10^{16} E_{g0}[\text{eV}]$.

$$w = \frac{2^8 E_{g0}}{\pi\hbar} \int \sum_{n,m} |f_{nm}|^2 \delta(1 + \xi^2 + \varepsilon_1^2/2 + \varepsilon_2^2/2 - \omega_1 n - \omega_2 m) d^3 \xi \approx \\ \approx 1.24 \cdot 10^{17} E_{g0}[\text{eV}] \int \sum_{n,m} |f_{nm}|^2 \delta(1 + \xi^2 + \varepsilon_1^2/2 + \varepsilon_2^2/2 - \omega_1 n - \omega_2 m) d^3 \xi$$

For 9 eV and density of atoms $\rho_0 = 6.6 \cdot 10^{22} \text{ cm}^{-3}$ the coefficient to multiply keldformax.dat $= W_{PI0}[\text{s}^{-1}] \rho_0 = 3.677 \cdot 10^{40} \text{ sm}^{-3} \text{ s}^{-1}$.

Commented [TD23]: You mean cm⁻³, maybe

According to calculation for 9 eV 800 nm integral is $\approx 1.92 \cdot 10^{-3}$, thus $w \approx 1.07 \cdot 10^{13} \text{ s}^{-1}$. For density of atoms $6.6 \cdot 10^{22}$ coefficient is $7 \cdot 10^{35} \text{ cm}^{-3} \text{ s}^{-1}$.

$$\frac{\partial \psi}{\partial t} = i\Delta \psi - i\{\omega \varepsilon, r\}(\psi + \psi_0) \\ \dot{a}_k - \{\omega \varepsilon, \nabla_k\} a_k = -ik^2 a_k + ia_0 e^{it} \{\omega \varepsilon, \nabla_k\} V_{0k} \quad (3) \\ V_{0k} = \frac{8\pi^{1/2}}{(1+k^2)^2}$$

Solving (3) in an usual way with characteristics in k space with zero initial data we have

$$a_{\xi} = e^{-i \int_0^t (\xi - \mathbf{q}(\tau))^2 d\tau} \int_0^t \{\omega \varepsilon(\tau), (\xi - \mathbf{q}(\tau))\} W(|\xi - \mathbf{q}(\tau)|) e^{i\tau + i \int_0^{\tau} (\xi - \mathbf{q}(v))^2 dv} d\tau \quad (4)$$

$$W(k) = -(1 + \mathbf{k}^2)^{-3},$$

Atom, after saddle point.

In Keldysh's work as well as in mine, formulas (16) and (18) are obtained after calculating the integral value with saddle-point method. Those formulas can be handled better after rewriting them as (16) contains misprint: instead of $\sqrt{2I_0/\hbar}$ expression $\sqrt{2I_0}/(\hbar\omega)$ had to be written)

$$w_0 = \omega \sqrt{2x_0} \left(\frac{\gamma}{\sqrt{1+\gamma^2}} \right)^{3/2} \exp \left[-2x \left(\ln \left(\gamma + \sqrt{1+\gamma^2} \right) - \gamma \frac{\sqrt{1+\gamma^2}}{1+2\gamma^2} \right) \right].$$

$$\cdot \exp \left[-2 \left(\langle x+1 \rangle - x \right) \left(\ln \left(\gamma + \sqrt{1+\gamma^2} \right) - \frac{\gamma}{\sqrt{1+\gamma^2}} \right) \right].$$

$$\cdot \sum_{n=0}^{\infty} \Phi(z) \exp \left[-2n \left(\ln \left(\gamma + \sqrt{1+\gamma^2} \right) - \frac{\gamma}{\sqrt{1+\gamma^2}} \right) \right]$$

$$z = \left[\frac{2\gamma}{\sqrt{1+\gamma^2}} \left(\langle x+1 \rangle - x + n \right) \right]^{1/2}$$

$$x_0 = \frac{I_0}{\hbar \omega}$$

$$x = x_0 (1 + \varepsilon^2 / 2)$$

$$\gamma = \varepsilon^{-1}$$

$$\Phi(z) = \int_0^z e^{y^2 - z^2} dy$$

Commented [TD24]: Is it related to a parabolic band structure?

Or

$$w_0 = \omega \sqrt{2x_0} \left(\frac{\gamma}{\sqrt{1+\gamma^2}} \right)^{3/2} \left(\frac{\varepsilon}{2} \right)^{2\langle x+1 \rangle} \exp \left[\frac{-2x\gamma^3}{(1+2\gamma^2)\sqrt{1+\gamma^2}} - 2 \left(\langle x+1 \rangle \left(\ln \left(\frac{\gamma + \sqrt{1+\gamma^2}}{2\gamma} \right) - \frac{\gamma}{\sqrt{1+\gamma^2}} \right) \right) \right].$$

$$\cdot \sum_{n=0}^{\infty} \Phi(z) \left(\frac{\varepsilon}{2} \right)^{2n} \exp \left[-2n \left(\ln \left(\frac{\gamma + \sqrt{1+\gamma^2}}{2\gamma} \right) - \frac{\gamma}{\sqrt{1+\gamma^2}} \right) \right]$$

The radiation wavelength is much longer than size of the atom. Thus, the electrical field depends only on time and is defined as follows:

$$E = E_1(t) \mathbf{e}_1 \cos(\omega_1 t) + E_2(t) \mathbf{e}_2 \cos(\omega_2 t + \varphi_2),$$

Commented [TD25]: Same comment as before: E(t) = E1 e1 cos(...) + E2 e2 cos(...)

Where E_1, E_2 states for first and second harmonic pulse, which are polarized along constant vectors $\mathbf{e}_1, \mathbf{e}_2$. Some of following calculations can be performed for time-dependent amplitudes. Thus, time dependence of E_1, E_2 is still considered. Harmonics can be shifted by phase φ_2 . The main results are obtained for $E_{1,2} = \text{const.}$ In [1] the case of $E_1 = \text{const.}, E_2 = 0$ is considered.

To make a clear overview, all values are normalized as follows. Distance will be measured in terms of $a_0 = \hbar^2 / (m_e \alpha)$ and time in $\hbar / E_{g0}$, where $E_{g0} = \frac{m_e \alpha^2}{2\hbar^2}$ is a ground state energy; frequency will be measured in $E_{g0} / \hbar$, electrical field in Keldysh's units $E_{*1,2} = \omega_{1,2} (2m_e E_{g0})^{1/2} e^{-1}$. Thus, the electrical field of each component is normalized on its own value. Then the dimensionless quantities $\varepsilon_{1,2} = E_{1,2} / E_{*1,2}$ will be used instead of electrical fields. Hence, the Schrödinger equation for the continuous spectrum function ψ takes the form:

$$\frac{\partial \psi}{\partial t} = i\Delta \psi - i\{\omega \varepsilon, r\}(\psi + \psi_0) \quad (1)$$

$$\{\omega \varepsilon, r\} = \omega_1 \varepsilon_1(t)(\mathbf{e}_1 \mathbf{r}) \cos(\omega_1 t) + \omega_2 \varepsilon_2(t)(\mathbf{e}_2 \mathbf{r}) \cos(\omega_2 t + \varphi_2), \quad (2)$$

Where $\psi_0 = a_0 e^{i\mathbf{r} \cdot \mathbf{r}}$ is the wave function of ground state. The factor a_0 is insignificant, since coefficients of obtained speed of ionization values will be determined correspondingly to an experiment. It is only important that constant a_0 is independent on parameters of the electromagnetic wave, mainly on its frequency. Function $\{\omega \varepsilon, r\} \psi_0$ doesn't have component along ψ_0 in space of functions, since $\int \psi_0^* \{\omega \varepsilon, r\} \psi_0 d^3 r = 0$. This fact was taken into account in (1). After Fourier transform $\psi = \int a_k e^{i\mathbf{k} \cdot \mathbf{r}} d^3 k$ in (1) we get

$$\dot{a}_k - \{\omega \varepsilon, \nabla_k\} a_k = -ik^2 a_k + ia_0 e^{it} \{\omega \varepsilon, \nabla_k\} V_{0k} \quad (3)$$

$$V_{0k} = \frac{8\pi^{1/2}}{(1+k^2)^2}$$

After solving (3) in an usual way with characteristics in k space with zero initial data we have

$$a_\xi = e^{-i \int_0^t (\xi - \mathbf{q}(\tau))^2 d\tau} \int_0^t \{\omega \varepsilon(\tau), (\xi - \mathbf{q}(\tau))\} W(|\xi - \mathbf{q}(\tau)|) e^{i\tau + i \int_0^\tau (\xi - \mathbf{q}(\nu))^2 d\nu} d\tau \quad (4)$$

$$W(k) = -(1+k^2)^{-3}, \quad (5)$$

$$\mathbf{q}(t) = \mathbf{e}_1 \int_0^t \omega_1 \varepsilon_1(\tau) \cos(\omega_1 \tau) d\tau + \mathbf{e}_2 \int_0^t \omega_2 \varepsilon_2(\tau) \cos(\omega_2 \tau + \varphi_2) d\tau$$

Since our main interest is dependence of $\int |a_k|^2 d^3 k$, then factors of type $a_0 e^{i\beta}$, ($\beta \in \text{Re}$) in front of integral (4) are not important. Integral over k in infinite limits is equal to integral over ξ . Then we have

Commented [TD26]: Same question: what is alpha? Which unit does it has? Maybe this text is repeated?

$$a_{\xi} = \int_0^t \frac{\omega_1 \varepsilon_1(\tau) \cos(\omega_1 \tau) (\mathbf{e}_1 \cdot (\xi - \mathbf{q}(\tau)))}{(1 + (\xi - \mathbf{q}(\tau))^2)^3} e^{i\tau + i \int_0^{\tau} (\xi - \mathbf{q}(\nu))^2 d\nu} d\tau +$$

$$+ \int_0^t \frac{\omega_2 \varepsilon_2(\tau) \cos(\omega_2 \tau + \varphi_2) (\mathbf{e}_2 \cdot (\xi - \mathbf{q}(\tau)))}{(1 + (\xi - \mathbf{q}(\tau))^2)^3} e^{i\tau + i \int_0^{\tau} (\xi - \mathbf{q}(\nu))^2 d\nu} d\tau$$

or

$$a_{\xi} = \int_0^t \left[\omega_1 \varepsilon_1(\tau) \cos(\omega_1 \tau) (\mathbf{e}_1 \cdot (\xi - \mathbf{q}(\tau))) + \omega_2 \varepsilon_2(\tau) \cos(\omega_2 \tau + \varphi_2) (\mathbf{e}_2 \cdot (\xi - \mathbf{q}(\tau))) \right] \times$$

$$\times \frac{\exp \left(i\tau + i \int_0^{\tau} (\xi - \mathbf{q}(\nu))^2 d\nu \right)}{(1 + (\xi - \mathbf{q}(\tau))^2)^3} d\tau \quad (6)$$

In case of constant pulse

$$\mathbf{q}(t) = \varepsilon_1 \sin(\omega_1 t) \mathbf{e}_1 + \varepsilon_2 \sin(\omega_2 t + \varphi_2) \mathbf{e}_2 \quad (7)$$

Then (6) becomes

$$a_{\xi} = \int_0^t \left[\omega_1 \varepsilon_1 \cos(\omega_1 \tau) (\mathbf{e}_1 \cdot (\xi - \mathbf{q}(\tau))) + \omega_2 \varepsilon_2 \cos(\omega_2 \tau + \varphi_2) (\mathbf{e}_2 \cdot (\xi - \mathbf{q}(\tau))) \right] \times$$

$$\times \frac{\exp \left(i\tau + i \int_0^{\tau} (\xi - \mathbf{q}(\nu))^2 d\nu \right)}{(1 + (\xi - \mathbf{q}(\tau))^2)^3} d\tau$$

$$\tau + \int_0^{\tau} (\xi - \mathbf{q}(\nu))^2 d\nu = (1 + \xi^2) \tau + 2\varepsilon_1 \frac{\cos(\omega_1 \tau) - 1}{\omega_1} (\xi \cdot \mathbf{e}_1) + \varepsilon_2 \frac{\cos(\omega_2 \tau + \varphi_2) - 1}{\omega_2} (\xi \cdot \mathbf{e}_2) +$$

$$+ \varepsilon_1^2 \left(\frac{\tau}{2} - \frac{\sin(2\omega_1 \tau)}{4\omega_1} \right) + \varepsilon_2^2 \left(\frac{\tau}{2} - \frac{\sin(2(\omega_2 \tau + \varphi_2)) - \sin(2\varphi_2)}{4\omega_2} \right) +$$

$$+ \varepsilon_1 \varepsilon_2 (\mathbf{e}_1 \cdot \mathbf{e}_2) \left(\frac{\sin((\omega_1 - \omega_2)\tau - \varphi_2) + \sin(\varphi_2)}{\omega_1 - \omega_2} - \frac{\sin((\omega_1 + \omega_2)\tau + \varphi_2) - \sin(\varphi_2)}{\omega_1 + \omega_2} \right) = \quad (8)$$

$$= (1 + \xi^2 + \varepsilon_1^2 / 2 + \varepsilon_2^2 / 2) \tau + \eta + \text{const}$$

$$\eta(\tau) = 2\varepsilon_1 \frac{\cos(\omega_1 \tau)}{\omega_1} (\xi \cdot \mathbf{e}_1) + 2\varepsilon_2 \frac{\cos(\omega_2 \tau + \varphi_2)}{\omega_2} (\xi \cdot \mathbf{e}_2) +$$

$$-\varepsilon_1^2 \frac{\sin(2\omega_1\tau)}{4\omega_1} - \varepsilon_2^2 \frac{\sin(2(\omega_2\tau + \varphi_2))}{4\omega_2} + \varepsilon_1\varepsilon_2(\mathbf{e}_1 \cdot \mathbf{e}_2) \left(\frac{\sin((\omega_1 - \omega_2)\tau - \varphi_2)}{\omega_1 - \omega_2} - \frac{\sin((\omega_1 + \omega_2)\tau + \varphi_2)}{\omega_1 + \omega_2} \right) \quad (9)$$

Terms independent on time (const in (8)) in exponent can be separated from the integral over time in (6) and can be neglected. Denoting

$$f = \frac{\omega_1\varepsilon_1 \cos(\omega_1\tau)(\mathbf{e}_1 \cdot (\boldsymbol{\xi} - \mathbf{q}(\tau))) + \omega_2\varepsilon_2 \cos(\omega_2\tau + \varphi_2)(\mathbf{e}_2 \cdot (\boldsymbol{\xi} - \mathbf{q}(\tau)))}{(1 + (\boldsymbol{\xi} - \mathbf{q}(\tau))^2)^3} \quad (10)$$

Thereafter,

$$a_\xi = \int_0^t e^{i(1+\xi^2+\varepsilon_1^2/2+\varepsilon_2^2/2)\tau} f e^{i\eta(\tau)} d\tau$$

2. Calculation of ionization speed in case of irrational ratio ω_1 / ω_2 .

In case of an irrational relation ω_1 / ω_2 the function $f e^{i\eta}$ will be expanded into double Fourier series in terms of $e^{i\omega_1 t}$ and $e^{i\omega_2 t}$ as follows:

$$f e^{i\eta} = \sum_{m,n} f_{n,m} e^{-in\omega_1 t - im\omega_2 t}, \quad (11)$$

We note that functions f and η are periodical functions of variables $\omega_1 t$ and $\omega_2 t + \varphi_2$. Consider the function of two variables after replacing $\omega_1 t$ with $\omega_1 t_1$ and $\omega_2 t$ with $\omega_2 t_2$ in $f e^{i\eta}$ term:

$$F(\omega_1 t_1, \omega_2 t_2 + \varphi_2) = f(\omega_1 t_1, \omega_2 t_2 + \varphi_2) e^{i\eta(\omega_1 t_1, \omega_2 t_2 + \varphi_2)}.$$

Function $F(x, y)$ is periodic with period of 2π for both variables. We expand it into a double Fourier series as $F = \sum_{m,n} F_{n,m} e^{-in\omega_1 t_1 - im\omega_2 t_2}$. Obviously, $f_{n,m} = F_{n,m}$ and

$$f_{n,m} = \frac{\omega_1}{2\pi} \frac{\omega_2}{2\pi} \int_0^{2\pi/\omega_1} dt_1 \int_0^{2\pi/\omega_2} F(\omega_1 t_1, \omega_2 t_2 + \varphi_2) e^{in\omega_1 t_1 + im\omega_2 t_2} dt_2 \quad (12)$$

Then

$$a_\xi = \int_0^t f e^{i\eta} e^{i(1+\xi^2+\varepsilon_1^2/2+\varepsilon_2^2/2)\tau} d\tau$$

$$\int |a_\xi|^2 d^3 \xi = \int d^3 \xi \int_0^t d\mu \int_0^t d\nu f(\mu) f^*(\nu) e^{i(\eta(\mu) - \eta(\nu)) + i(1+\xi^2+\varepsilon_1^2/2+\varepsilon_2^2/2)(\mu - \nu)}$$

$$= \sum_{m,n,m',n'} \int d^3\xi f_{m,n} f_{m',n'}^* \int_0^t d\mu \int_0^t dv e^{i(1+\xi^2+\varepsilon_1^2/2+\varepsilon_2^2/2)(\mu-\nu)-i\omega_1 n \mu - i\omega_2 m \mu + i\omega_1 n' \nu + i\omega_2 m' \nu} \quad (13)$$

We define average ionization speed over long time as $W_{PI} = \lim_{t \rightarrow \infty} t^{-1} \int |a_\xi|^2 d^3\xi$. Considering usual limit representation of δ -functions as imaginary exponentials up to unimportant factors we see that only series terms with $n = n'$, $m = m'$ are left in (13) and

$$W_{PI} = \sum_{n,m} \int |f_{m,n}|^2 \delta(1+\xi^2+\varepsilon_1^2/2+\varepsilon_2^2/2-\omega_1 n - \omega_2 m) d^3\xi \quad (14)$$

It is obvious from (12) and (14) that phase φ_2 doesn't affect the results, thus its value can be set as 0. Indeed, replacing $\omega_2 t \rightarrow \omega_2 t + \varphi_2$ in (12) leads to multiplication of $f_{n,m}$ by $e^{-im\varphi_2}$, which doesn't affect $|f_{n,m}|$.

Using cylindrical coordinate system $\xi_\perp, \varphi, \xi_z$ with axis $\mathbf{e}_1$ and defining $\xi_{m,n}^2 = \omega_1 n + \omega_2 m - (1 + \varepsilon_1^2/2 + \varepsilon_2^2/2)$, from (13) we have

$$W_{PI} = W_{PI0} \sum_{n,m=-\infty}^{\infty} W_{m,n} \quad (15)$$

$$W_{m,n} \equiv \int |f_{m,n}|^2 \delta(1+\xi^2+\varepsilon_1^2/2+\varepsilon_2^2/2-\omega_1 n - \omega_2 m) d^3\xi = \quad (16)$$

$$= \int |f_{m,n}(\xi_\perp^2, \xi_z, \varphi)|^2 \delta(\xi_\perp^2 - (\xi_{m,n}^2 - \xi_z^2)) \xi_\perp d\xi_\perp d\xi_z d\varphi = \frac{1}{2} \int_{-\xi_{m,n}}^{\xi_{m,n}} \int_0^{2\pi} |f_{m,n}(\xi_{m,n}^2 - \xi_z^2, \xi_z, \varphi)|^2 d\xi_z d\varphi \quad (17)$$

Here, W_{PI0} is a numerical factor independent on wave parameters. This factor must be determined after comparison to experiment.

Delta function in (14) and (16) has clear physical meaning. The energy of n quanta from wave with frequency ω_1 and m quanta from a wave with frequency ω_2 is used to overcome ionization potential (equals to 1 in chosen units), given to a free electron by an average energy of oscillation $\varepsilon_1^2/2$ in first wave and $\varepsilon_2^2/2$ in second wave and the energy of translation movement ξ^2 . Value $W_{n,m}$ represents the speed of n, m -photon ionization. n, m -photon ionization is not possible, if the energy of corresponding amount of quanta is not sufficient for overcoming forbidden band (ionization potential) and providing average oscillation energy ($\omega_1 n + \omega_2 m < 1 + \varepsilon_1^2/2 + \varepsilon_2^2/2$). In other words, $W_{n,m} = 0$ in case of sufficiently high field amplitude. Dependency of ionization speed on wave amplitude shows a local decrease of W_{PI} (fig. 1).

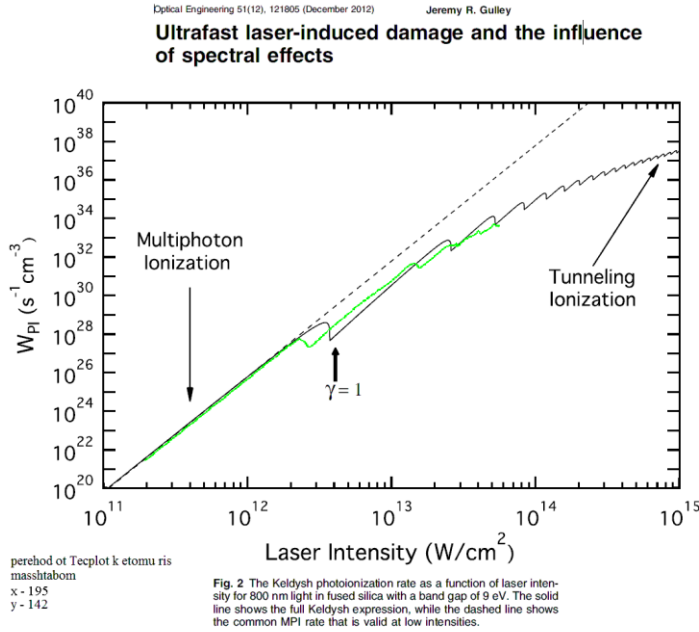


Fig. 1 Dependency of ionization speed on monochromatic laser wave intensity according to [14] and in current work. Details can be found in sec. 6.
 Thibault this is old version which is not good! Don't pay attention.

3. Calculation of ionization speed in case of rational ω_1 / ω_2 .

If for frequencies ω_1 and ω_2 exists a frequency ω and integers $n_{\omega 1}$ and $m_{\omega 2}$, so that $\omega_1 = n_{\omega 1} \omega$, $\omega_2 = m_{\omega 2} \omega$ (the fraction $m_{\omega 2} / n_{\omega 1}$ is considered as irreducible), then expansion $f e^{i\eta}$ into single Fourier series over frequency ω must be used instead of (11). In case expansion into double Fourier series (11) is used, first terms with same $n\omega_1 + m\omega_2$ must be summarized, and then the absolute value

must be summarized over all possible values $n\omega_1 + m\omega_2$: $\sum_L \left| \sum_{n,m: n\omega_1 + m\omega_2 = L\omega} f_{nm} \right|^2$. The

first method of ionization speed calculation is faster for small $n_{\omega 1}, m_{\omega 2}$. Otherwise the method using double Fourier series and summarizing with equal $n\omega_1 + m\omega_2$ is faster. In latter case pairs of numbers (n, m) and $(\bar{n}, \bar{m})$, which result into same number $n\omega_1 + m\omega_2 = \bar{n}\omega_1 + \bar{m}\omega_2$, are separated from each other by very large distance $(n - \bar{n})^2 + (m - \bar{m})^2 \gg 1$. Since W_{nm} noticeably differs from 0 in

restricted area of space (n, m) , formulas for irrational ω_1 / ω_2 can be used instead of mentioned summarization for large $n_{\omega_1}, m_{\omega_2}$.

Already for $\omega_2 / \omega_1 = 2/1$ the differences between calculation of W_{PI} using formulas for irrational frequencies ratio and formal usage of formulas for irrational frequencies ratio are low. (Tab. 1). We emphasize that the phase φ_2 doesn't play a role anymore in case of irrational frequency ratio. However already for frequency ratio 2/1 moderate dependence on phase can be observed (Tab.1). With further increase of n_{ω_1} and m_{ω_2} the mentioned dependence disappears.

Tab. 1. $E_{g0} = 9$ eV, $\lambda_1 = 0.8$ μm , $\lambda_2 = 0.4$ μm , $\varepsilon_1^2 = \varepsilon_2^2 = 0.45$. A and Б – calculation by formulas rational frequencies ratio for $\varphi_2 = 0$ and $\varphi_2 = \pi/2$ respectively; B – formal use of formulas for irrational ratio; Γ – case of $\lambda_1 = 0.8$ μm , $\lambda_2 = 0.41$ μm . Dependence on phase φ_2 is not present. Result is equal to one obtained with formal use of formulas for irrational ω_2 / ω_1 .

		W_{PI}
$\mathbf{e}_1 \perp \mathbf{e}_2$	A	$.20993 \cdot 10^{-3}$
	Б	$.21367 \cdot 10^{-3}$
	B	$.212655 \cdot 10^{-3}$
	Γ	$.195 \cdot 10^{-3}$
$\mathbf{e}_1 \parallel \mathbf{e}_2$	A	$.300 \cdot 10^{-3}$
	Б	$.33941 \cdot 10^{-3}$
	B	$.315 \cdot 10^{-3}$
	Γ	$.299 \cdot 10^{-3}$

The case of equal frequencies $\omega_1 = \omega_2$, $\mathbf{e}_1 \perp \mathbf{e}_2$ and $\varphi_2 = 0$ is equivalent to a case $\varepsilon_2 = 0$ (linearly polarized monochromatic pulse) and only differs by polarization direction. The case of $\mathbf{e}_1 \perp \mathbf{e}_2$, $\omega_1 = \omega_2$ and $\varphi_2 = \pi/2$ corresponds to circular polarization. Calculations show that for the same wave energy (same value $\varepsilon_1^2 + \varepsilon_2^2$) the speed of ionization is bigger in case of linearly polarized waves than for circularly polarized waves. This fact corresponds to experimental data. For example, for wavelength 0.8 μm , $E_{g0} = 9$ eV, $\varepsilon_1^2 = \varepsilon_2^2 = 0.45$ ratio of W_{PI} for a linearly polarized wave compared to W_{PI} of a circularly polarized wave is 1.24/0.234.

We note that in [11] the analogical qualitative dependencies of ionization speed on wave polarization for substantially different pulse parameters were obtained.

We further note that a real pulse has a width. Thus, it might be more correct to the formulas for irrational frequencies ration.

4. Energy absorbed during ionization

In case of ω_1 / ω_2 is irrational, the energy absorbed from first and second wave is $Q_1 = \sum_{n,m} n\omega_1 W_{m,n}$ and $Q_2 = \sum_{n,m} m\omega_2 W_{m,n}$ respectively, based on above-mentioned physical meaning of (14) and (16). In dimensional form, one multiplies by E_{g0} ($n\omega \rightarrow n\hbar\omega$). In case of ω_1 / ω_2 is rational, the total energy absorbed from both waves $Q = \sum_{n,m} (n\omega_1 + m\omega_2) W_{m,n}$ can be easily calculated. However, it is not known how this energy is distributed between waves. Considering remarks made in chapter 3, formulas for irrational frequency ratio may be apparently used.

We will discuss some details of the model for describing the interaction between two color laser pulses with glasses. For a pulse length in the order of tens of femtoseconds it is suitable to use complex amplitudes to represent electric field, current and other quantities. For example, real electric field can be represented as $\sum_{\alpha=1,2} (E_\alpha e^{-i\omega_\alpha t} + E_\alpha^* e^{i\omega_\alpha t}) / 2$ (α is number of a wave in pulse). This representation is necessary, since in many formulas of the model (for example, for Kerr effect) the complex amplitudes are used. Additionally, it simplifies the problem of numerical equation solving, since time of E_α change is significantly longer then ω_α^{-1} . Maxwell equation (in CGS) for electric field will be as follows:

$$\frac{1}{c} \frac{\partial \mathbf{D}_\alpha}{\partial t} - i \frac{\omega_\alpha}{c} \mathbf{D}_\alpha = \frac{4\pi}{c} e \rho \mathbf{v}_\alpha + \text{rot } \mathbf{B}_\alpha - \frac{8\pi}{c} G_\alpha \frac{\mathbf{E}_\alpha}{|\mathbf{E}_\alpha|^2} \quad (18)$$

Where $\mathbf{D}_\alpha$, $\mathbf{v}_\alpha$ and $\mathbf{B}_\alpha$ are complex amplitudes of induction vector, speed of free electrons and magnetic field respectively, ρ is a density of free electron plasma, e is an elementary charge, G_α is energy absorbed during photoionization, α is number of wave in the pulse. For the electron velocity and density following equations are used:

$$\frac{\partial(\rho \mathbf{v}_\alpha)}{\partial t} - i\omega_\alpha \rho \mathbf{v}_\alpha = -\rho \frac{e}{m_e} \mathbf{E}_\alpha - \frac{\rho \mathbf{v}_\alpha}{\tau_c} \quad (19)$$

$$\partial \rho / \partial t = W_{pi} + W_\sigma \quad (20)$$

Where m_e is a mass of free electron, τ_c is the collision time with the lattice (phonons), W_σ is the rate of other ionization-recombination processes (for example, impact ionization). In hydrodynamic equation (19) the terms with electron pressure and $(\mathbf{v}\nabla)\mathbf{v}$ are omitted. Those terms are insignificant for our calculations.

In many cases, a simpler equation can be used instead of (19):

$$i\omega_\alpha \mathbf{v}_\alpha = \frac{e}{m_e} \mathbf{E}_\alpha + \mathbf{v}_\alpha / \tau_c \quad (21)$$

The last term in (18) is arranged in such a way that while deriving the energy conservation law for electromagnetic energy, G_α is subtracted from energy. Let us find the ratio between G_α and Q_α . We note that in case the variation of complex amplitudes is slow compared to wave frequency, equations (18) - (20) provide subtraction of average electron oscillation energy in a wave $\frac{|\mathbf{E}_\alpha|^2 e^2}{4\omega_\alpha^2 m_e}$ (in dimensionless units of previous chapters $\varepsilon_\alpha^2/2$) from energy of electromagnetic field. This subtracted energy is already taken into account in Keldysh's theory, in the expression for Q_α . Thus, in case of equations (18)-(20), $G_\alpha = Q_\alpha - W_{pl} \varepsilon_\alpha^2/2$ must be set. In equations (18) and (21) mentioned energy is not subtracted, since (21) corresponds to steady oscillations. Thus, $G_\alpha = Q_\alpha$ must be set.

We note a sharp increase of absorbed energy Q while transferring from multiphoton ionization regime with small wave amplitudes to tunnel regime. In fig.2, the amount of absorbed quanta per electron $Q/(W_{pl} \hbar \omega)$ versus intensity $I = cE^2/(8\pi)$ is shown. In analogy to the article [14] the intensity of a light wave (for a given electric field in a wave) in vacuum, rather than in glass is understood under the term intensity. The case of fused silica ($E_{g0} = 9$ eV) with electron (polaron) mass equal to half of electron mass in vacuum m_{vac} and monochromatic wave at 800nm is introduced.

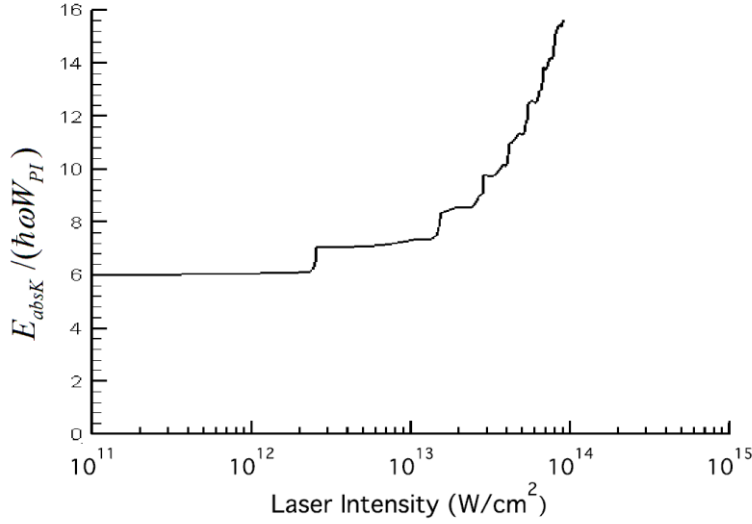


Fig. 2

We also note that for fused silica and a wavelength of 800 nm, at minimum 6 quants are needed for ionization. However, for such low intensities, that there is no glass modification anymore, the frequently used equation for 6 photon ionization $W_{PI} \sim I^6$ stops to be valid.

5. Integration method while calculating ionization rate

In [1] integration is carried out by the saddle-point method. In current work, numerical methods for integrals and sums in (12), (15) and (17). Such calculation requires significant computational resources. Range of m and n, used during summation, is chosen in a way that the result does not change significantly with further increase of series terms. To evaluate fast oscillating integrals of form

$$I = \frac{\omega}{2\pi} \int_0^{2\pi/\omega} F(t) e^{i\eta(t) + i\alpha\omega t} dt$$

The following method was used. The integration range is divided in N_t segments $t_k = k\tau$, $k = 0, \dots, N_t$, $\tau = 2\pi/(N_t\omega)$. In segments $t_{k-1} < t < t_{k+1}$ F is approximated by quadratic polynom:

$$F(t) = F_k + \frac{F_{k+1} - F_{k-1}}{2\tau} (t - t_k) + \frac{F_{k+1} - 2F_k + F_{k-1}}{2\tau^2} (t - t_k)^2$$

The exponent is represented in the form

$$\eta_k + n\omega t_k + \left(\frac{\eta_{k+1} - \eta_{k-1}}{2\tau} + n\omega \right) (t - t_k)$$

Then

$$I = \frac{\omega}{2\pi} \sum_{k=1, N_1-1, 2} e^{i\eta_k + i\omega n t_k} \int_{-\tau}^{\tau} \left(f_k + \frac{f_{k+1} - f_{k-1}}{2\tau} \sigma + \frac{f_{k+1} - 2f_k + f_{k-1}}{2\tau^2} \sigma^2 \right) e^{i(n + \nu_k)\omega \sigma} d\sigma$$

where $\nu_k = \frac{\eta_{k+1} - \eta_{k-1}}{2\tau\omega}$. Integration in those expressions can be easily carried out analytically.

Integration over φ and ξ_z was made with rectangle method ($\int_{-a}^a g(x) dx \approx \sum_{j=1, J_0} g(x_{i+1/2}) \frac{2a}{J_0}$, $x_{i+1/2} = -a + 2a \frac{j-1/2}{J_0}$) or Simpson's rule. The number of grid points was selected experimentally. Calculations show that Simpson's rule is more suitable for integrating over ξ_z . 30 grid nodes are sufficient to bring error below 1%. In many cases, rectangular method converges faster than Simpson's rule while integrating over φ . Besides, the result doesn't change at 10-s decimal place with further increase of J_0 already for small amount of grid nodes $J_0 \sim 15$. A possible explanation of this can be as follows.

The quantity f_{nm} is a function of $\sin \varphi$ and a $|f_{nm}|^2$ is a function of $\sin^2 \varphi$. The latter statement is based on some amount of calculations, but hasn't been proved analytically. Correspondingly, the quantity $\int_{-\xi_{m,n}}^{\xi_{m,n}} |f_{m,n}(\xi_{m,n}^2 - \xi_z^2, \xi_z, \varphi)|^2 d\xi_z$ is expanded into a series of the form $\sum_{k=0, \infty} g_k \cos(2k\varphi) d\varphi$, and

$I = \int_{-\pi/2}^{\pi/2} \int_{-\xi_{m,n}}^{\xi_{m,n}} |f_{m,n}(\xi_{m,n}^2 - \xi_z^2, \xi_z, \varphi)|^2 d\xi_z d\varphi = \int_{-\pi/2}^{\pi/2} \sum_{k=0, \infty} g_k \cos(2k\varphi) d\varphi = \pi g_0$. Rectangular method gives $I \approx \pi g_0 + \pi \sum_{L=1, \infty} (-1)^{L(J_0+1)} g_{LJ_0}$. Thus, the error of rectangular method

is defined by high number harmonics. The error of calculation of I by Simpson's rule is defined by g_k with $k < J_0$, multiplied by a grid step in the power of 4. The better convergence of rectangle method might be due to the fact that g_k decreases rapidly (for example, exponentially or as k^{-1} to the high power) with increase of k .

For completeness, we note that $|f_{nm}|^2$, are apparently the even functions of ξ_z . This is also not proven analytically. In order to avoid errors, calculations were carried out over whole range of φ and ξ_z , without taking possible symmetry into account.

6. Difference between crystal (semiconductor) and atom in Keldysh theory.

The theory of photoionization for atom and crystal is developed in [1]. The difference between those cases is basically only in a dispersion law. For a crystal, energy depends on an electron-hole pair pulse p as follows: [1]

$$\varepsilon(p) = E_{g0} \sqrt{1 + p^2 / (mE_{g0})} \approx E_{g0} + p^2 / (2m) - p^4 / (8m^2 E_{g0}) \quad (22)$$

While for atom holds $\varepsilon(p) = E_{g0} + p^2 / (2m_e)$, where m_e is electron mass in medium, $m = m_e m_h / (m_e + m_h)$ is a reduced mass of electron and hole. In the low-intensity limit, dispersion law for a crystal becomes an expression for an atom, but with reduced mass. In case of a crystal, the expressions similar to (11)-(17) for ionization rate appear. There, $\langle \varepsilon(p) \rangle$ (average over period) figures instead of average oscillation energy ε_1^2 , where the impulse is determined from equation $dp/dt = -eE \cos(\omega t)$.

Fused silica is an amorphous body. Therefore, Keldysh's photoionization theory for an atom is more suitable than theory of photoionization in a crystal. Nevertheless, we investigate differences between atom and crystal for a monochromatic wave for completeness.

Fig. 1 shows ionization rate versus wave intensity in vacuum for a crystal from [14] and calculated by formulas from current work. In [14] $m = 0.5m_{\text{evac}}$ (m_{evac} is electron mass in vacuum). $m_e = 0.5m_{\text{evac}}$ is set in (11)-(17) to achieve a good agreement with the case of amorphous body in a low intensity limit. The corresponding coefficient W_{p10} is adjusted as well.

$\langle \varepsilon(p) \rangle$ increases slower than $\langle p^2 / (2m) \rangle$ in (22). Hence, for such intensities, for which n-photon ionization finishes in case [14], ionization rate drops later. However, one must note that dependencies are not only qualitatively similar, but as well quantitatively. Considering imperfections of Keldysh's theory while describing fused silica (see below) as well as uncertainties of formulas describing other processes during laser pulse and glass interaction (impact ionization, the frequency of collisions of free electrons and atoms etc.), those differences are not significant. Additionally, dispersion relation (22) is as well imperfect [15-18].

We note one more difficulty in use of (22) for describing femtosecond pulse with matter interaction with equations of type (18) - (21). In this case carrier velocity in expression for flux will be nonlinearly related to impulse:

Commented [TD27]: Okay, we can find this coefficient using my TDDFT simulations.

$v = \partial \varepsilon / \partial p \neq p/m$. This leads to harmonics generation and difficulties with carrier frequency separation.

Thus, the following can be proposed. While developing the model of femtosecond pulse interaction with dielectrics, the ~~polaron-electron~~ mass contained in W_{pl} should be considered as a parameter. That parameter should be adjusted after comparing to an experiment.

7. Calculation results

Some important results of W_{pl} calculation:

For large wave amplitudes, when $\varepsilon_1^2/2 + \varepsilon_2^2/2 > \omega_1 n + \omega_2 m - 1$, corresponding ionization becomes impossible ($W_{m,n} = 0$). On plots of ionization rate versus wave amplitude (fig.1) a local decrease of W_{pl} can be observed. For fused silica that is important for practice (forbidden band $E_{g0} = 9$ eV) and wavelength 800 nm (monochromatic pulse) the smallest n at which $W_n \neq 0$ is equal to 6. However, according to experiment, 6-photon ionization condition $\varepsilon_1^2/2 < 6\omega_1 - 1$ stops to be valid for amplitudes too small to cause glass modification. Thus, frequently used formula for multiphoton ionization $W_{pl} \sim I^6$ is only applicable as rough guess. The sums (15) with large amount of terms must be used. We also note that absorbed wave of electromagnetic wave per electron also increases with wave amplitude (fig.2) Thus, it is incorrect to use constant value for absorbed energy per electron (for example $6\hbar\omega$ or E_{g0}).

Calculations show that the rate of ionization by two color pulse is mostly significantly higher than total rate of ionization by two separated pulses of same amplitude: $W_{pl}(\varepsilon_1, \varepsilon_2) > W_{pl}(\varepsilon_1, \varepsilon_2 = 0) + W_{pl}(\varepsilon_1 = 0, \varepsilon_2)$ (fig.3). The values $\varepsilon_1, \varepsilon_2$ for which W_{pl} decreases can be exception ($\varepsilon_1^2/2 + \varepsilon_2^2/2 \approx \omega_1 n + \omega_2 m - 1$, $W_{nm} \rightarrow 0$).

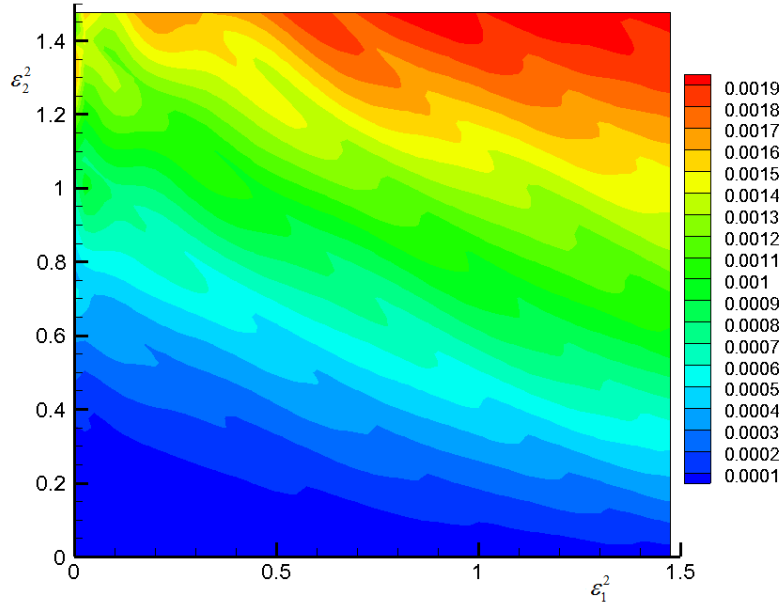


Fig. 3. W_{pi} versus squared dimensionless wave amplitudes in two color pulse in case of irrational frequencies ratio. Wavelengths 800 and 401 nm. $\mathbf{e}_1 \parallel \mathbf{e}_2$

Ionization rate in case of two color pulse with same polarization waves ($\mathbf{e}_1 \parallel \mathbf{e}_2$) is higher than in case of perpendicular polarizations $\mathbf{e}_1 \perp \mathbf{e}_2$. Particularly, this was demonstrated for monochromatic wave with circular polarization that can be considered as two color pulse with $\mathbf{e}_1 \perp \mathbf{e}_2$, same frequencies and phase $\varphi_2 = \pi/2$.

At high fields it can happen that energy of one wave is enough for ionization and second wave for amplification. For example, in case of waves with same polarization ($\mathbf{e}_1 \parallel \mathbf{e}_2$) with $\lambda_1 = 800$ and $\lambda_2 = 401$ nm, dimensionless amplitudes $\varepsilon_1 = 1.8$, $\varepsilon_2 = 0.1$ first wave loss is $3.4E_{g0}$ per electron. The second wave obtains energy $0.0136E_{g0}$. Those results are obtained for irrational frequency ratio. We note that the use of code for rational frequency ratio in this case results into same ionization rate. However, one cannot distribute pulse energy loss between waves. Distribution of W_{nm} in this case is shown at fig. 4. For $\varepsilon_2 = 0$ value $W_{nm} \neq 0$ is only for $m=0$, and for $\varepsilon_1 = 0$ – for $n=0$.

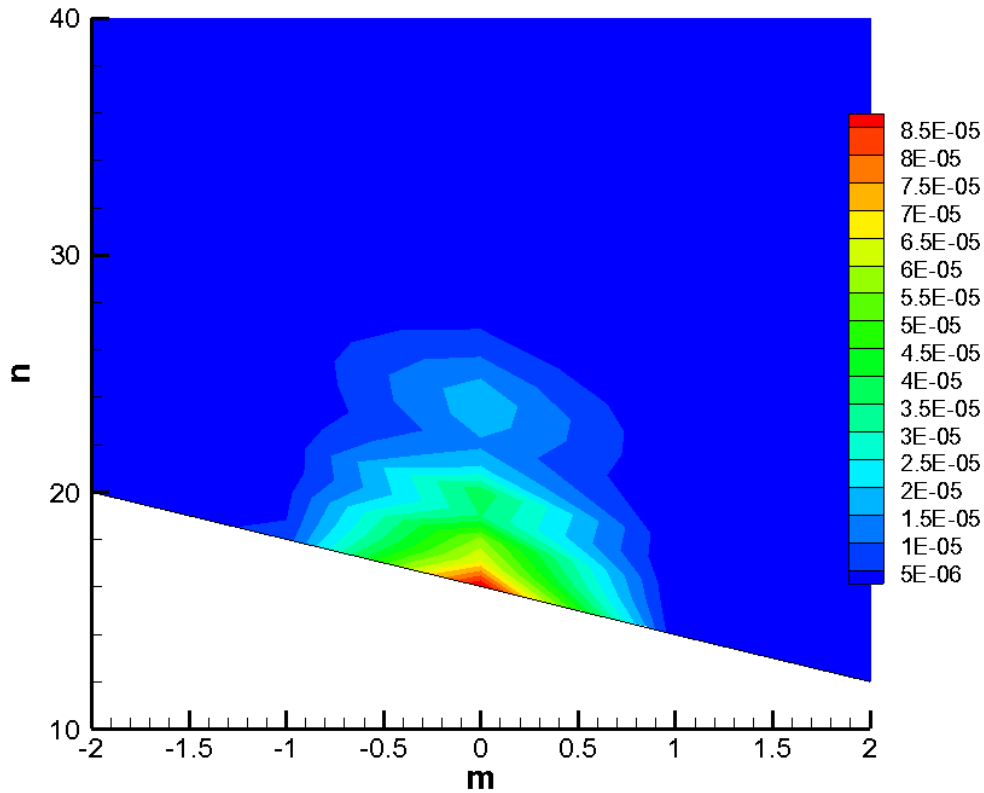


Fig. 4 W_{nm} distribution. Pulse parameters are listed in chapter 7.

8. Model disadvantages

The Keldysh model of photoionization and its generalization, developed in current work has following disadvantages:

1. Free electron for a long period (compared to a wave period) is considered as unaffected except of by mentioned wave. This is not valid in case of fused silica. Collision frequency is comparable to wave period. However, for big amount of collisions, impact ionization starts to play a significant role and photoionization is not significant.
2. Time for amplitude change must be much higher than wave period. For pulse length around tens of femtoseconds, amplitude change in a given position can happen much faster (<10 fs) thanks to fast change of plasma density. **Thus, wave period value is lying in short range.**
3. Dependence of conducting electron impulse on its energy can differ from proposal in [1]. This can significantly affect the result. [Gruzdev].

To some extent, those disadvantages can be leveled by the fact that in many practically important cases photoionization acts as fuse. Most of conducting electrons are produced after impact ionization. We note that while describing impact ionization many problems arise. Thus, the model developed in current work and original model in [1] must be considered as semi phenomenological model and included coefficients as parameters which must be determined after comparison to an experiment.

9. Conclusion

The Keldysh theory for photoionization in a solid body (dielectric, semiconductor) is generalized for a two-color laser pulse in current work. Many interesting and important results are obtained and explained in chapter 8.

We emphasize the complexity, variety and interrelation of processes behind laser and matter interaction. As a result, equations describing different processes during interaction are connected to each other. Particularly, current work shows that the term describing electron absorption due to photoionization in Maxwell's equation depends on the type of equation used for conduction electron velocity. Technically, dispersion law (relation between energy and impulse of conducting electron) used in photoionization theory must correspond to equation for electrons movement in hydrodynamic part of model for Maxwell's or Schrödinger's nonlinear equations. The fully correct mathematical description of investigated effects is very complicated, obviously not possible and maybe even not needed. One must create simple semi phenomenological models that consider basic process properties with parameters determined after comparison to experiment. Results of current work are first of all relevant for fused silica, but may be also applicable to other materials.

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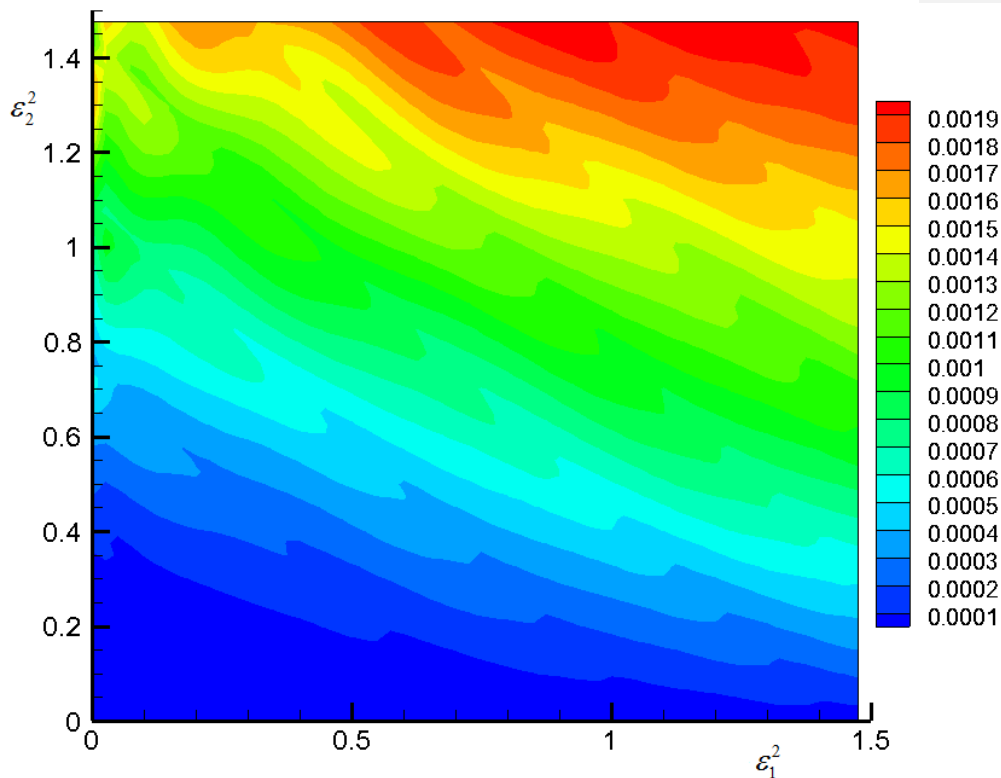


Figure. 1. W_{pl} as a function of squared dimensionless wave amplitude in two color pulse in case of irrational frequencies ration. Wavelengths are 800 and 401 nm.

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